5-MINUTE REVISION

Missing Plot Analysis

Handling a lost observation in CRD, RCBD and Latin square: direct fitting, Bartlett's covariance method and substitution.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · Chapter revision

KAU Me-LON · www.agrinoteskau.com · Dept. of Agricultural Statistics, College of Agriculture, Vellayani

THE IDEA TO KEEP

Estimate a missing plot; do not treat it as a new observation.

REMEMBER

Use the method appropriate to the design and reduce error degrees of freedom for estimated values.

APPLY IT

An RBD with one missing plot loses one error degree of freedom relative to the corresponding complete-data analysis.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 Why plots go missing

Animal or pest damage, flooding, theft, recording errors... Always **record why** an observation is missing.

A poor result **caused by a treatment** is real data and must not be discarded.

A missing plot **destroys the balance** of an RCBD or Latin square; a CRD copes directly.

2 CRD: fit directly

Analyse the observed responses with unequal treatment sample sizes (no substitution needed).

Error df

$$edf = n_{obs} - v$$

SE of a difference

$$SE = \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

3 RCBD: one missing value

Substitution estimate

$$\hat{x} = \frac{rB + vT - G}{(r-1)(v-1)}$$

B, T = observed totals of the affected block and treatment; G = observed grand total (all **excluding** the missing value).

Error df

$$(r-1)(v-1) - 1$$

4 Latin square: one missing value

Substitution estimate

$$\hat{x} = \frac{v(R + C + T) - 2G}{(v-1)(v-2)}$$

R, C, T = observed totals of the affected row, column and treatment; G = observed grand total.

Error df

$$(v-1)(v-2) - 1$$

5 Bartlett's covariance method

Create an imaginary **covariate** X: 1 (or −1) for the missing plot, 0 elsewhere. Set $Y = 0$ for the missing plot. Then run **ANCOVA** for the design.

The indicator accounts for the missing position: total and error df are each **one less** than in the complete design.

6 After substituting

- ✓ The estimate **minimises the error SS**.
- ✓ Subtract **one** error df per estimated value.
- ✓ The treatment SS can be **inflated**: apply the bias correction.
- ✓ Comparisons involving the affected treatment need adjusted SEs.
- ✓ Several missing values: estimate by **iteration**.

7 Choosing an approach

Design	Approach	Error df
CRD	Fit observed data (unequal n_i)	$n_{obs} - v$
RCBD	Substitution $\hat{x}$, Bartlett's method, or fit the observed data	$(r-1)(v-1)-1$
Latin square	Substitution $\hat{x}$ or Bartlett's method	$(v-1)(v-2)-1$

! Common mistakes

- ✗ Counting $\hat{x}$ as a real observation.
- ✗ Including the missing cell in B, T or G.
- ✗ Ignoring missingness linked to the response itself (bias).

? Fill in the blanks

1. In an RBD with r = 4 and v = 5, B = 60, T = 50 and G = 400 give a missing value estimate $\hat{x}$ = ——. **[7.5]**
2. An RBD with 4 treatments in 5 blocks and one missing plot has —— error df. **[11]**
3. A 4 × 4 Latin square with one missing plot has —— error df. **[5]**
4. The substituted value is the one that makes the error sum of squares ——. **[as small as possible]**
5. In a CRD, a lost plot is handled by analysing with —— replications. **[unequal]**
6. In Bartlett's method the dummy covariate is —— for the missing plot and 0 elsewhere. **[1]**

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